SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

**INNOVATION BY DESIGN 2025**
Presented by: **MONDAY - BROWN**

## PROBLEM STATEMENT

**"Travelers struggle to express themselves when overseas due to language barriers."**

**They encounter confusion, delays, and misunderstandings in conversations because they cannot convey meaning accurately through speech and language.**

**How Might We help travelers to communicate effectively.**

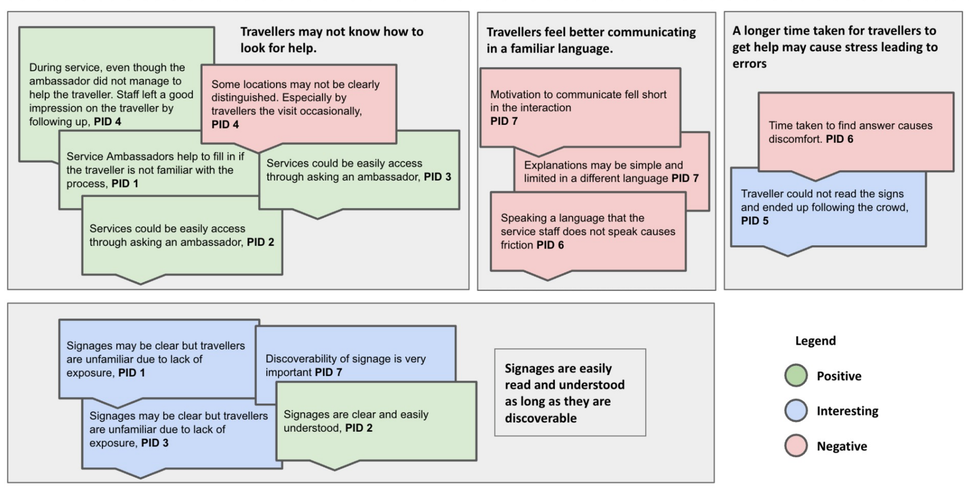
## CONTEXTUAL INQUIRY + INTERVIEW

**10**
Solo Travelers

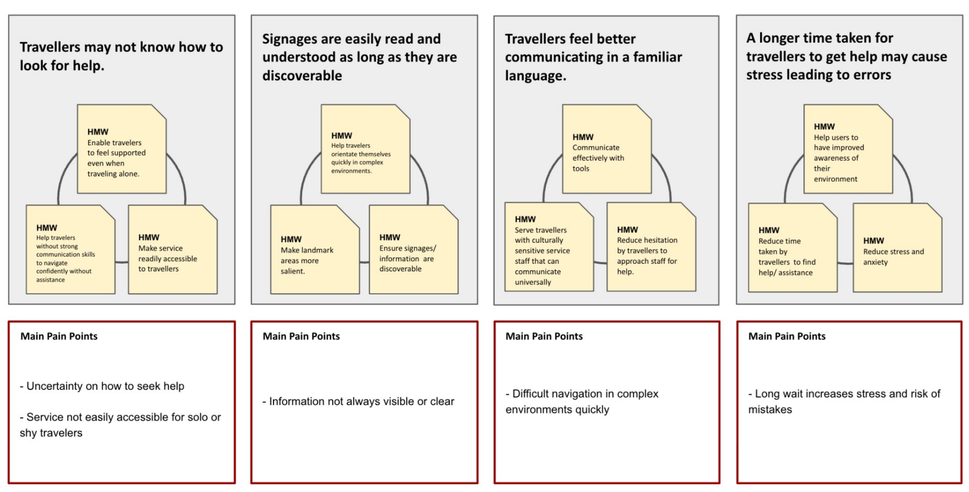
**6**
Students

**4**
Travelers

## AFFINITY ANALYSIS (VERBATIM)




## PAIN POINTS




## HOW MIGHT WE...

Communicate without using language.

Reduce the time taken to align ourselves in a conversation

Improve the experience of communication between two languages

## PRODUCT FEATURES STUDY

**Methods Used:**

Mind Map | Product Concept Evaluation | Pugh Comparison Matrix

**Insight**

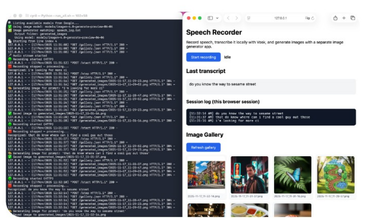
- **Camera**, **AR Display**, and **AI System** have the highest total score (S=7), indicating these features stand out as highly suitable for further design consideration despite mixed individual ratings for assembly difficulty and development risk. Integrating these would produce the most valuable design.

**Evaluation**

- Based on scores, the system design should **explore integrating a camera vision, AR display, and AI system components.** (if possible)
- Features with a score of 6 (**microphone, microcomputer**) are valid options to include explore.

## PRODUCT DEVELOPMENT

> To design a device that can turn speech into meaningful graphics



System Prototyping + Iterative Design

**INSIGHT #1 - Images are also abstract and may not serve its purpose.**
**( HMW )** Provide other forms of system feedback for user.

User Testing

**INSIGHT #2 - Processing of image may be delayed. Not useful to user if it takes too long.**
**( HMW )** Automate user inputs to make interaction seamless.

Final Design

---

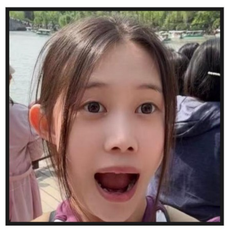

**MASTER OF SCIENCE IN TECHNOLOGY AND DESIGN**

**Hu** Yangruotao

**LIU** Lyu

**WANG** Yuting

**SHI** Cynthia



**TANG** Cyril